

First Order Logic (Predicate Calculus)

This Lecture

Last time we talked about propositional logic, a logic on simple statements.

This time we will talk about first order logic, a logic on quantified statements.

First order logic is much more expressive than propositional logic.

The topics on first order logic are:

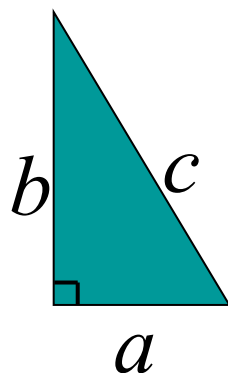
- Quantifiers
- Negation
- Multiple quantifiers
- Arguments of quantified statements

Limitation of Propositional Logic

Propositional logic – logic of simple statements

$$\neg, \wedge, \vee, \rightarrow, \leftrightarrow$$

How to formulate Pythagoreans' theorem using propositional logic?



How to formulate the statement that there are infinitely many primes?

Predicates

Predicates are propositions (i.e. sentences) with variables
And becomes a statement when specific values are substituted for the variables

Example: $P(x,y) ::= x + 2 = y$

$x = 1$ and $y = 3$: $P(1,3)$ is true

$x = 1$ and $y = 4$: $P(1,4)$ is false
 $\neg P(1,4)$ is true

More Examples: $P(x) ::= x$ is a student at BUET

$Q(x,y) ::= x$ is a student at y

When there is a variable, we need to specify what to put in the variables.

The **domain** of a variable is the set of all values that may be substituted in place of the variable.

Set

To specify the domain, we often need the concept of a **set**.
Roughly speaking, a set is just a collection of objects.

Some examples

R	Set of all real numbers
Z	Set of all integers
Q	Set of all rational numbers

Given a set, the (only) important question is whether an element belongs to it.

$x \in A$ means that x is an **element** of A (pronounce: x in A)

$x \notin A$ means that x is **not** an **element** of A (pronounce: x not in A)

Sets can be defined explicitly: e.g. $\{1,2,4,8,16,32,\dots\}$, $\{\text{CSE103},\text{CSE105},\dots\}$

Truth Set

Sometimes it is inconvenient or impossible to define a set explicitly.

Sets can be defined by a predicate

Given a predicate $P(x)$ and x has domain D , the **truth set** of $P(x)$ is the set of all elements of D that make $P(x)$ true.

$$\{x \in D \mid P(x)\}$$

e.g. Let $P(x)$ be "x is the square of a number",
and the domain D of x is the set of positive integers.

Then the truth set is the set of all positive integers which are the square of a number.

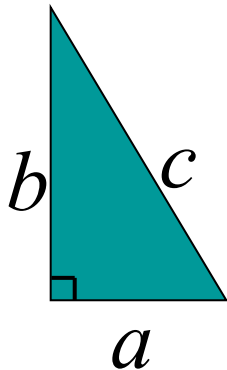
e.g. Let $P(x)$ be "x is a prime number",
and the domain D of x is the set of positive integers.

Then the truth set is the set of all positive integers which are prime numbers.

The Universal Quantifier

The universal quantifier $\forall$ for ALL x

Example: $\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}, x + y = y + x.$



Pythagorean's theorem

$\forall$ right – angled triangle If a, b, c are ... then
$$a^2 + b^2 = c^2$$

Example: $\forall x \quad x^2 \geq x$ This statement is true if the domain is $\mathbb{Z}$ (integers), but not true if the domain is $\mathbb{R}$ (reals).

The truth of a predicate depends on the domain.

Truth and Falsity of Universal Statements

- Let $D = \{1,2,3,4,5\}$ and consider the statement: $\forall x \in D, x^2 \geq x$. Show this is true.

Method of Exhaustion

$1^2 = 1 \geq 1$	$2^2 = 4 \geq 2$	$3^2 = 9 \geq 3$	$4^2 = 16 \geq 4$	$5^2 = 25 \geq 5$
TRUE	TRUE	TRUE	TRUE	TRUE

- $\forall x \in \mathbb{R}, x^2 \geq x$. Show that this is false.
 - Take $x = \frac{1}{2}$. Clearly $x \in \mathbb{R}$
 - $(\frac{1}{2})^2 = \frac{1}{4} < \frac{1}{2}$
 - So, this is false

Counterexample

The Existential Quantifier

$\exists$ There **EXISTS** some y

y.g. $\exists y, y^2 = y$

The truth of a predicate depends on the domain.

$$\forall x \exists y. x < y$$

Domain

Truth value

integers $\mathbb{Z}$

T

positive integers $\mathbb{Z}^+$

T

negative integers $\mathbb{Z}^-$

F

negative reals $\mathbb{R}^-$

T

Check -1.

Truth and Falsity of Existential Statements

- $\exists m \in \mathbb{Z}, m^2 = m$. Show that this is true.
 - Take $m = 1$. Clearly $m \in \mathbb{Z}$
 - $(1)^2 = 1$, So, this is true (at least for one element)
- Let $E = \{5, 6, 7, 8, 9\}$ and consider the statement: $\exists m \in \mathbb{Z}, m^2 = m$. Show this is false.

Method of Exhaustion				
$5^2=25 \neq 5$	$6^2=36 \neq 6$	$7^2=49 \neq 7$	$8^2=64 \neq 8$	$9^2=81 \neq 9$
FALSE	FALSE	FALSE	FALSE	FALSE

Implicit Quantification

- If a number is an integer, then it is a rational number
 - No quantification (for all/each/any)
 - But the presence of indefinite article 'a' before the number indicates universal quantification.
 - $\forall x, x \in \mathbb{Z} \rightarrow x \in \mathbb{Q}$
- The number 24 can be written as a sum of two even integers
 - No quantification (there exists a/some)
 - $\exists$ even integers m and n such that $24 = m+n$

Translating Mathematical Theorem

Fermat (1637): If an integer n is greater than 2,
then the equation $a^n + b^n = c^n$ has no solutions in non-zero integers a , b , and c .

$$\forall n > 2 \quad \forall a \in \mathbb{Z}^+ \quad \forall b \in \mathbb{Z}^+ \quad \forall c \in \mathbb{Z}^+ \quad a^n + b^n \neq c^n$$

Translating Mathematical Theorem

Goldbach's conjecture: Every even number is the sum of two prime numbers.

Suppose we have a predicate $\text{prime}(x)$ to determine if x is a prime number.

$$\forall n \in \mathbb{Z} \text{ even}(n) \rightarrow$$

$$\exists p \in \mathbb{Z} \exists q \in \mathbb{Z} \text{ prime}(p) \wedge \text{prime}(q) \wedge (p+q = n)$$

How to write $\text{prime}(p)$?

$$\text{prime}(p) :=$$

$$(p > 1) \wedge (\forall a \in \mathbb{Z} \forall b \in \mathbb{Z} ((a > 1) \wedge (b > 1) \rightarrow a \cdot b \neq p))$$

Exercises

- $\forall x \in \mathbb{R}, x^2 \geq 0$
 - All real numbers have nonnegative squares.
 - Every real number has a nonnegative square.
 - Any real number has a nonnegative square.
 - x has a nonnegative square, for each real number x .
 - The square of any real number is nonnegative.
- $\forall x \in \mathbb{R}, x^2 \neq -1$
 - All real numbers have squares not equal to -1.
 - No real numbers have squares equal to -1.
- $\exists m \in \mathbb{Z}, m^2 = m$
 - There is an integer whose square is equal to itself.
 - We can find at least one integer equal to its own square.
 - $m^2 = m$, for some integer m .
 - Some integer equals its own square.
 - Some integers equal their own squares.

Exercises

- All triangles have three sides.
 - $\forall$ triangles t , t has three sides
 - $\forall t \in T$, t has three sides (where T is the set of all triangles).
- No dogs have wings.
 - $\forall$ dogs d , d does not have wings
 - $\forall d \in D$, d does not have wings (where D is the set of all dogs)
- Some programs are structured.
 - $\exists$ a program p such that p is structured
 - $\exists p \in P$ such that p is structured (where P is the set of all programs)

Equivalent Forms

- All primes greater than 2 are odd
 - $\forall x((\text{Prime}(x) \wedge x > 2) \rightarrow \text{Odd}(x))$
 - $\forall x \in D \text{ Odd}(x)$
 - Here D consists of all values of the variable x such that $x > 2$ and $\text{Prime}(x)$ is true.
- $\forall x \in U, \text{ if } P(x) \text{ then } Q(x)$

Equivalently we can say:

- $\forall x \in D, Q(x);$
 - Here D consists of all values of the variable x that makes $P(x)$ true.

- Quantifiers
- Negation
- Multiple quantifiers
- Arguments of quantified statements

Negations of Quantified Statements

Everyone likes football.

What is the negation of this statement?

~~No one likes football.~~



Not (everyone likes football) = There exists someone who doesn't like football.

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$



(generalized) DeMorgan's Law

Say the domain has only three values.

$$\begin{aligned}\neg \forall x P(x) &\equiv \neg (P(1) \wedge P(2) \wedge P(3)) \\ &\equiv \neg (P(1) \wedge P(2)) \vee \neg P(3) \\ &\equiv \neg P(1) \vee \neg P(2) \vee \neg P(3) \equiv \exists x \neg P(x)\end{aligned}$$

The same idea can be used to prove it for any number of variables, by mathematical induction.

Negations of Quantified Statements

There is a plant that can fly.

What is the negation of this statement?

~~There is a plant
that cannot fly.~~

 **WRONG**

Not (exists a plant that can fly) = every plant cannot fly.



$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

(generalized) DeMorgan's Law

Say the domain has only three values.

$$\begin{aligned}\neg \exists x P(x) &\equiv \neg (P(1) \vee P(2) \vee P(3)) \\ &\equiv \neg (P(1) \vee P(2)) \wedge \neg P(3) \\ &\equiv \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \\ &\equiv \forall x \neg P(x)\end{aligned}$$

The same idea can be used to prove it for any number of variables, by mathematical induction.

Negation Exercises

- $\forall$ primes p , p is odd.
 - $\exists$ a prime p such that p is not odd.
- $\exists$ a triangle T such that the sum of the angles of T equals 200°
 - $\forall$ triangles T , the sum of the angles of T does not equal 200°

Informal -> Formal -> Negate

- No politicians are honest
 - Formal: $\forall$ politicians x , x is not honest.
 - Negation: $\exists$ a politician x such that x is honest.
 - Informal Negation: Some politicians are honest

Informal Negation: Ambiguity

- All mathematicians wear glasses

Informally Negate:

Depends on the
verbal stress

- All mathematicians do not wear glasses

- Meaning 1: There are mathematicians who do not wear glasses.
- Meaning 2: None of the mathematicians wears glasses.

WRONG

Negation of Universal Implications

$$\begin{aligned} & \neg (\forall x, P(x) \rightarrow Q(x)) \quad \neg \forall x P(x) \equiv \exists x \neg P(x) \\ & \equiv \exists x \text{ such that } \neg (P(x) \rightarrow Q(x)) \\ & \equiv \exists x \text{ such that } \neg (\neg P(x) \vee Q(x)) \\ & \equiv \exists x \text{ such that } (P(x) \wedge \neg Q(x)) \end{aligned}$$

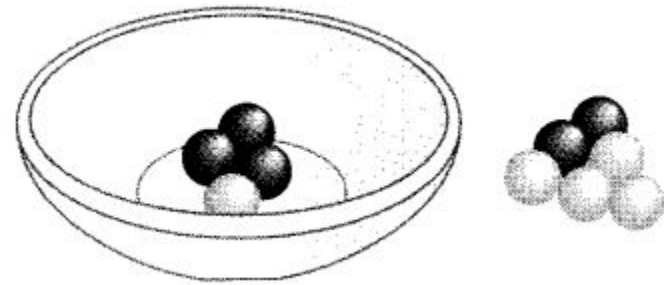
Example negation:

- $\forall$ people p , if p is blond then p has blue eyes
- $\exists$ a person p such that, p is blond and p does not have blue eyes

Vacuous Truth

We have 5 black and 5 white balls and a bowl

TRUE/FALSE?



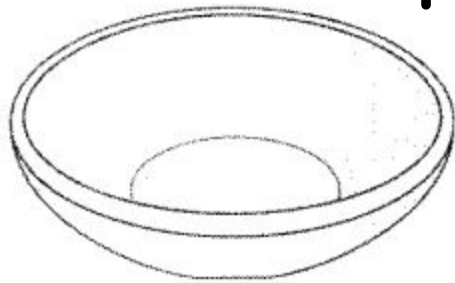
All the balls in the bowl are black

FALSE

$P =$ All the balls in the bowl are black. 

~~**FALSE**~~

TRUE



$\neg P =$

Negation:

There exists a ball in the bowl that is not black. **FALSE**

Is there a ball in the bowl that is non-black, i.e., white? **NO**

P is FALSE if and only if $\neg P$ is TRUE

Vacuous Truth

- $\forall x \in D, P(x) \rightarrow Q(x)$ is vacuously true or true by default if and only if $\forall x \in D, \neg P(x)$.

Contrapositive, Converse, Inverse

- $\forall x \in D, P(x) \rightarrow Q(x)$
- Contrapositive
 - $\forall x \in D, \neg Q(x) \rightarrow \neg P(x)$
- Converse
 - $\forall x \in D, Q(x) \rightarrow P(x)$
- Inverse
 - $\forall x \in D, \neg P(x) \rightarrow \neg Q(x)$

Contrapositive, Converse, Inverse

- If a real number is greater than 2, then its square is greater than 4.
- $\forall x \in \mathbb{R}, x > 2 \rightarrow x^2 > 4$
- Contrapositive
 - $\forall x \in \mathbb{R}, x^2 \not> 4 \rightarrow x \not> 2$
- Converse
 - $\forall x \in \mathbb{D}, x^2 > 4 \rightarrow x > 2$
- Inverse
 - $\forall x \in \mathbb{D}, x \not> 2 \rightarrow x^2 \not> 4$

Necessity, Sufficiency, if, Only if

“ $\forall x, r(x)$ is a **sufficient condition** for $s(x)$ ”

$\equiv$ “ $\forall x, \text{if } r(x) \text{ then } s(x).$ ”

“ $\forall x, r(x)$ is a **necessary condition** for $s(x)$ ”

$\equiv$ “ $\forall x, \text{if } \sim r(x) \text{ then } \sim s(x)$ ”

$\equiv$ “ $\forall x, \text{if } s(x) \text{ then } r(x).$ ”

} Contrapositive

“ $\forall x, r(x)$ **only if** $s(x)$ ”

$\equiv$ “ $\forall x, \text{if } \sim s(x) \text{ then } \sim r(x)$ ”

$\equiv$ “ $\forall x, \text{if } r(x) \text{ then } s(x).$ ”

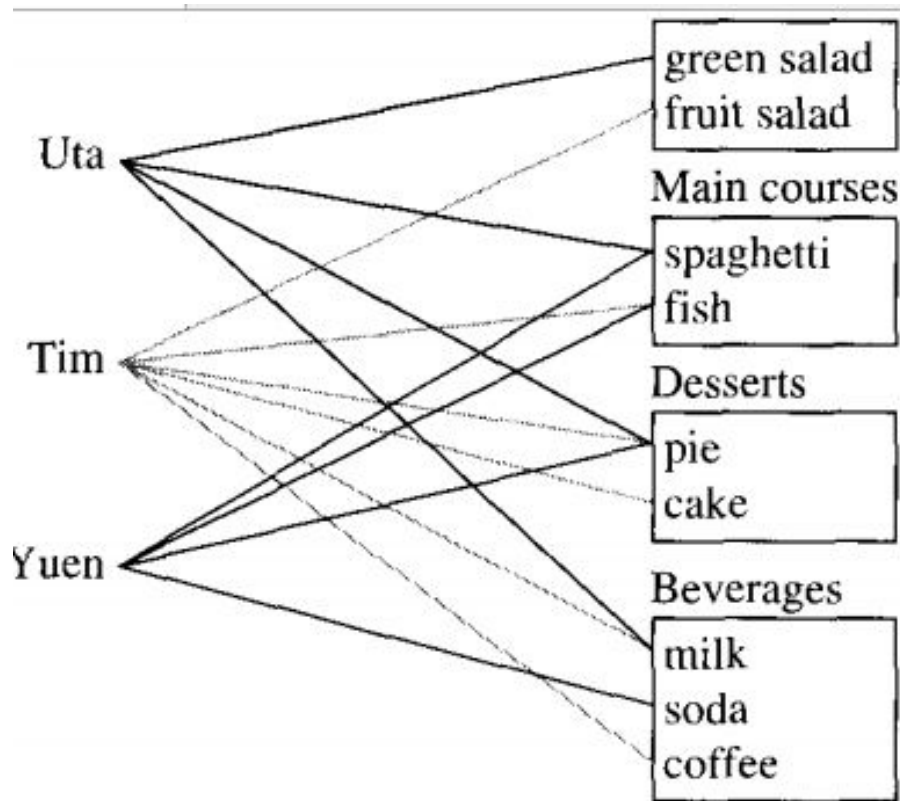
} Contrapositive

- Quantifiers
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Double Quantifiers

- **$\forall x \in D, \exists y \in E$ such that $P(x, y)$**
- How to establish the truth of the above?
 - For each element x in D
 - find an element y in E such that $P(x, y)$
- **$\exists x \in D$, such that $\forall y \in E, P(x, y)$**
- How to establish the truth of the above?
 - Find an element x in D for which each element y in E gives $P(x, y)$

Exercises: Formal & truth value



1. There is an item that was chosen by every student.
 - This is true; every student chose pie.
2. There is a student who chose every available item.
 - This is false; no student chose all nine items.
3. There is a student who chose at least one item from every station.
 - This is true; both Uta and Tim chose at least one item from every station.
4. Every student chose at least one item from every station.
 - This is false; Yuen did not choose a salad.

1. $\exists$ an item I such that $\forall$ students S , S chose I .
2. $\exists$ a student S such that $\forall$ items I , S chose I .
3. $\exists$ a student S such that $\forall$ stations Z , $\exists$ an item I in Z such that S chose I .
4. $\forall$ students S and $\forall$ stations Z $\exists$ an item I in Z such that S chose I .

Order of Quantifiers

There is an anti-virus program killing every computer virus.

How to interpret this sentence?

For every computer virus, there is an anti-virus program that kills it.

There is one (single) anti-virus program that kills all computer viruses.

Order of Quantifiers

There is an anti-virus program killing every computer virus.

How to interpret this sentence?

For every computer virus, there is an anti-virus program that kills it.

$$\forall V \exists P, \text{kill}(P, V)$$

- For every attack, I have a defense:
- against **MYDOOM**, use Defender
- against **ILOVEYOU**, use Norton
- against **BABLAS**, use Zonealarm ...

□□ is expensive!

Order of Quantifiers

There is an anti-virus program killing every computer virus.

How to interpret this sentence?

There is one single anti-virus program that kills all computer viruses.

$$\exists P \forall V, \text{kill}(P, V)$$

I have *one* defense good against every attack.

Example: P is CSE-antivirus,
protects against *ALL* viruses

That's much better!

Order of quantifiers is very important!

Order of Quantifiers

There is a person supervising every detail of the production process.

How to interpret this sentence?

There is one single person who supervises all the details of the production process.

$\exists p$ such that $\forall pp$ supervise(p, pp)

Order of Quantifiers

There is a person supervising every detail of the production process.

How to interpret this sentence?

For any particular production detail, there is a person who supervises that detail
(but there might be different supervisors for different details.)

$\forall pp, \exists p$ such that $\text{supervise}(p, pp)$

Order of Quantifiers

There is a person supervising every detail of the production process.

How to interpret this sentence?

$\exists p$ such that $\forall pp$ supervise(p, pp)

$\forall pp, \exists p$ such that supervise(p, pp)

In real life we have this sort of ambiguous situation and depending on the context we realize which one is actually intended for.

$$\neg(\forall x \forall y P(x,y)) = \exists x \neg(\forall y P(x,y)) = \exists x \exists y \neg$$

$$\neg(\forall x \exists y P(x,y)) = \exists x \neg(\exists y P(x,y)) = \exists x \forall y \neg$$

$$\neg(\exists x \forall y P(x,y)) = \forall x \neg(\forall y P(x,y)) = \forall x \exists y \neg$$

$$\neg(\exists x \exists y P(x,y)) = \forall x \neg(\exists y P(x,y)) = \forall x \forall y \neg$$

TABLE 1 Quantifications of Two Variables.

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

More on Negations

There is an anti-virus program killing every computer virus.

$$\exists P \forall V, \text{kill}(P, V)$$

What is the negation of this sentence?

$$\neg(\exists P \forall V, \text{kill}(P, V))$$

$$\equiv \forall P \neg(\forall V, \text{kill}(P, V))$$

$$\equiv \forall P \exists V \neg \text{kill}(P, V)$$

For every program, there is some virus that it can not kill.

Exercises

1. There is a smallest positive integer.
2. There is no smallest positive real number.
3. There are infinitely many prime numbers.

Exercises

1. There is a smallest positive integer.

$$\exists s \in \mathbb{Z}^+ \forall x \in \mathbb{Z}^+ s \leq x$$

$$\begin{aligned} & \neg(\exists s \in \mathbb{Z}^+ \forall x \in \mathbb{Z}^+ s \leq x) \\ & \equiv \forall s \in \mathbb{Z}^+ \neg(\forall x \in \mathbb{Z}^+ s \leq x) \\ & \equiv \forall s \in \mathbb{Z}^+ \exists x \in \mathbb{Z}^+ \neg(s \leq x) \\ & \equiv \forall s \in \mathbb{Z}^+ \exists x \in \mathbb{Z}^+ s \not\leq x \\ & \equiv \forall s \in \mathbb{Z}^+ \exists x \in \mathbb{Z}^+ x < s \end{aligned}$$

2. There is no smallest positive real number.

$$\forall r \in \mathbb{R}^+ \exists x \in \mathbb{R}^+ x < r$$

3. There are infinitely many prime numbers.

$$\forall p \in \mathbb{Z} \exists q \in \mathbb{Z} \text{prime}(p) \wedge \text{prime}(q) \wedge (q > p)$$

- Quantifiers
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Predicate Calculus Validity

Propositional validity

$$(A \rightarrow B) \vee (B \rightarrow A)$$

True *no matter what* the truth values of A and B are

Predicate calculus validity

$$\forall z [Q(z) \sqcap P(z)] \rightarrow [\forall x.Q(x) \sqcap \forall y.P(y)]$$

True *no matter what*

- the Domain is,
- or the predicates are.

That is, logically correct, independent of the specific content.

Predicate Calculus Validity

Propositional validity

$$(A \rightarrow B) \vee (B \rightarrow A)$$

True *no matter what* the truth values of A and B are

A	B	$A \rightarrow B$	$B \rightarrow A$	$(A \rightarrow B) \vee (B \rightarrow A)$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	1	1	1

Tautology

Predicate Calculus Validity

Predicate calculus validity

$$\forall z [Q(z) \supset P(z)] \rightarrow [\forall x.Q(x) \supset \forall y.P(y)]$$

True no matter what

- the Domain is,
- or the predicates are.

Similar to Tautology
Of Propositional logic

That is, logically correct, independent of the specific content.

- If hypothesis is false, the whole statement is true
- What happens, if the hypothesis is true?
 - In this case, for each Z, Q(Z) and P(Z) must be individually true.
 - So, the conclusion follows
 - So the statement is true

Arguments with Quantified Statements

Universal instantiation:

$$\begin{array}{l} \forall x, P(x) \\ \therefore P(a) \end{array}$$

Universal modus ponens:

$$\begin{array}{l} \forall x, P(x) \rightarrow Q(x) \\ P(a) \\ \therefore Q(a) \end{array}$$

Universal modus tollens:

$$\begin{array}{l} \forall x, P(x) \rightarrow Q(x) \\ \neg Q(a) \\ \therefore \neg P(a) \end{array}$$

Arguments with Quantified Statements

Universal instantiation:

$$\forall x, P(x) \\ \therefore P(a)$$

All men are mortal.
Socrates is a man.
 $\therefore$ Socrates is mortal.

- Simplify $r^{k+1} \times r$ 1 & UI
- $r^{k+1} \times r = \overbrace{r^{k+1} \times r^1} = r^{k+1+1} = r^{k+2}$
2 & UI

• We know from math:

1. For all real numbers x and all integers m and n , $x^m \times x^n = x^{m+n}$
2. For all real numbers x , $x = x^1$.

Arguments with Quantified Statements

Universal modus ponens: $\forall x, P(x) \rightarrow Q(x)$

$$P(a)$$

$$\therefore Q(a)$$

If a number is even, then its square is even.

k is a particular number that is even.

(We could say: 6 is even, i.e., $k=6$)

$\therefore k^2$ is even.

($\therefore 6^2 = 36$ is even)

$\forall x$, if $\boxed{x \text{ is even}}$ then $\boxed{x^2 \text{ is even.}}$

$$\forall x, E(x) \rightarrow S(x)$$

$$E(6)$$

$$\therefore S(6)$$

Formally

Arguments with Quantified Statements

Universal modus tollens: $\forall x, P(x) \rightarrow Q(x)$
 $\neg Q(a)$
 $\therefore \neg P(a)$

All human beings are mortal.
Zeus is not mortal.
 $\therefore$ Zeus is not human.

	$H(x)$		$M(x)$	
$\forall x$, if	x is human	then	x is mortal.	} Formally
$\forall x$, $H(x) \rightarrow$	$M(x)$			
$\neg M(\text{Zeus})$				
$\therefore \neg H(\text{Zeus})$				

All professors are absent-minded

Sohel Rahman is not absent-minded

Conclusion:???

Sohel Rahman is not a professor.

$\forall x \text{ professor}(x) \rightarrow$
 $\text{absent-minded}(x)$
 $\sim \text{absent-minded}(\text{Sohel})$
 $\therefore \sim \text{professor}(\text{Sohel})$

All criminals frequently go to that room

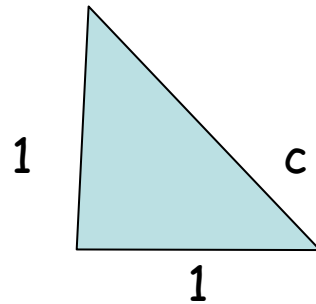
Sohel frequently goes to that room

Conclusion: Sohel is a criminal

$\forall x \text{ criminal}(x) \rightarrow$
 $\text{room}(x)$
 $\text{room}(\text{Sohel})$
 $\therefore \text{criminal}(\text{Sohel})$

Calculate c.

Are you using any argument we taught?



Valid Argument?

NO

Universal Generalization

valid rule

$$\frac{R(c)}{\forall x R(x)}$$

Informally, if we could prove that $R(c)$ is true for an arbitrary c (in a sense, c is a "variable"), then we could prove the for all statement.

e.g. given any number c , $2c$ is an even number

$\Rightarrow$ for all x , $2x$ is an even number.

Remark: Universal generalization is often difficult to prove, we will introduce mathematical induction to prove the validity of for all statements.

Argument Validity

- To say that an *argument form* is **valid** means the following:
 - No matter what particular predicates are substituted for the predicate symbols in its premises, if the resulting premise statements are all true, then the conclusion is also true.
- An *argument* is called valid if, and only if, its form is valid

Recall from Proposition...

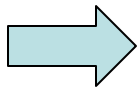
Argument

An argument is a sequence of statements.

All statements but the final one are called **assumptions** or **hypothesis**.

The final statement is called the **conclusion**.

An argument is **valid** if:



whenever all the assumptions are true, then the conclusion is true.

If today is Wednesday, then yesterday is Tuesday.

Today is Wednesday.

∴ Yesterday is Tuesday.

Valid Rule?

$$\forall z [Q(z) \rightarrow P(z)] \rightarrow [\forall x.Q(x) \rightarrow \forall y.P(y)]$$

Proof: Give **countermodel**, where
 $\forall z [Q(z) \rightarrow P(z)]$ is **true**,
but $\forall x.Q(x) \rightarrow \forall y.P(y)$ is **false**.

Find a domain,
and a predicate.

In this example, let domain be integers,

$Q(z)$ be true if z is an even number, i.e. $Q(z)=\text{even}(z)$

$P(z)$ be true if z is an odd number, i.e. $P(z)=\text{odd}(z)$

Then $\forall z [Q(z) \rightarrow P(z)]$ is true, because every number is either even or odd.

But $\forall x.Q(x)$ is not true, since not every number is an even number.

Similarly $\forall y.P(y)$ is not true, and so $\forall x.Q(x) \rightarrow \forall y.P(y)$ is not true.

Valid Rule?

$$\forall z \in D [Q(z) \supset P(z)] \rightarrow [\forall x \in D Q(x) \supset \forall y \in D P(y)]$$

Proof: Assume $\forall z [Q(z) \supset P(z)]$.

So $Q(z) \supset P(z)$ holds for all z in the domain D .

Now let c be some element in the domain D .

So $Q(c) \supset P(c)$ holds (by instantiation), and therefore $Q(c)$ by itself holds.

But c could have been any element of the domain D .

So we conclude $\forall x.Q(x)$. (by generalization)

We conclude $\forall y.P(y)$ similarly (by generalization). Therefore,

$$\forall x.Q(x) \supset \forall y.P(y) \quad \text{QED.}$$

Summary

This finishes the introduction to logic, half of the first part.

In the other half we will use logic to do mathematical proofs.

At this point, you should be able to:

- Express (quantified) statements using logic formula
- Use simple logic rules (e.g. DeMorgan, contrapositive, etc)
- Fluent with arguments and logical equivalence